

Europe's Green Deal at the Crossroads

ETS Carbon Pricing, Industrial Decarbonisation, and the Competitiveness Cost

Blue Lotus Research Intelligence | August 2026

Executive Summary

The European Union remains the world's most institutionally advanced climate actor, but its policy architecture is under growing structural strain. The EU Emissions Trading System (ETS) and its companion Carbon Border Adjustment Mechanism (CBAM) represent the centrepiece of the Green Deal's market-based approach, yet evidence from mid-2026 points to limited effectiveness at scale. The ETS has reduced emissions in covered sectors but its impact has been diluted by free allocation provisions; CBAM has been proposed as the corrective but faces protectionism accusations from trading partners and WTO compatibility questions. On the industrial side, steel and cement decarbonisation via hydrogen and carbon capture remains technically feasible but commercially nascent — the HYBRIT project in Sweden is among the most advanced real-world pilots. Europe's climate adaptation agenda must simultaneously contend with rising flood risk, heat stress, and drought, all of which are projected to intensify. Geopolitically, the EU's position as a climate frontrunner is increasingly tested by security pressures, rightward political drift, and the aftershock of US withdrawal from multilateral climate frameworks.

1. ETS Reform and the Carbon Price Architecture

The EU created its Emissions Trading System as the primary instrument to reduce pollution in high-emitting industries, including steel and chemicals. The scheme operates as a cap-and-trade mechanism: companies must purchase permits for each tonne of CO₂ they emit. Carbon removal credits in the broader voluntary market remain a rounding error in terms of volume — as of recent years, sales amounted to roughly 57 million tonnes, a marginal figure relative to total industrial emissions FT[2026-02-26].

The ETS has achieved measurable reductions in covered sectors, but its overall impact has been diluted by "free allocation" provisions that shield incumbent industries FT[2026-06-24]. Rising permit prices have created financial incentives to decarbonise, but the transition has been uneven. With energy prices soaring — in part due to supply disruptions — the financial calculus for green investment has grown more complex FT[2026-05-12]. In May 2026, the EU was actively considering extending the ETS to cover carbon costs on outbound international flights, a move that previously triggered retaliation from third-country carriers when first proposed for aviation FT[2026-05-12]. Extending the scheme to short-haul operators such as Ryanair formed part of those internal deliberations.

The revision of the ETS under the "Fit for 55" package — the EU's legislative framework for achieving its 2030 emissions reduction target — includes expansion of the trading system alongside updates to transport and buildings regulations. The Fit for 55 package also incorporates REPowerEU measures to

accelerate the EU's transition away from fossil fuels [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023-2024]. These policy packages represent a significant advance in scope but implementation gaps persist. Carbon uncertainty, flagged as the most acute risk factor for transition planning, has cast a shadow over investment decisions across the east of England and comparable industrial zones FT[2026-03-10].

2. Carbon Border Adjustment Mechanism: Design, Scope, and Geopolitical Friction

The CBAM is the EU's primary instrument for addressing carbon leakage — the risk that emissions-intensive production migrates to jurisdictions with less stringent regulation, eroding the environmental benefit of European decarbonisation unilaterally [RAG: CLIMATE_RAG — IPCC AR6 Chapter 11]. The absence of greenhouse gas pricing in competing jurisdictions can be interpreted as an implicit subsidy for fossil fuel-based production, providing an economic rationale for border carbon adjustment [RAG: CLIMATE_RAG — IPCC AR6 Chapter 11].

The EU formally proposed the CBAM regulation to cover materials already within the ETS scope, and a controversial proposed extension would bring in additional materials not yet covered FT[2025-11-10]. The mechanism is designed to shield EU heavy industry from the risk of production being displaced to cheaper, less-regulated regions. Several trade partners highly exposed to the EU market — including Turkey, South Korea, and the UK — have introduced carbon pricing or sought harmonisation with the EU system. However, the FT reported in June 2026 that the ETS and CBAM have "failed to create critical mass" in terms of price convergence internationally, with implementation remaining at limited scale FT[2026-06-24].

The UNDP Human Development Report 2023-24 identifies a structural tension in CBAM's design: if one bloc reduces emissions unilaterally, comparative advantage in greenhouse gas-intensive sectors shifts elsewhere — a phenomenon known as trade leakage. CBAM is the proposed corrective, but it may stimulate retaliatory conflict. The EU's earlier attempt to extend the ETS to international aviation drew retaliation from major states, a precedent that complicates CBAM expansion [RAG: UN_PRIMARY_RAG — UNDP Human Development Report 2023-24]. California remains the only sub-national jurisdiction to have implemented carbon border adjustment in practice [RAG: CLIMATE_RAG — IPCC AR6 Chapter 11].

3. Industrial Decarbonisation: Steel, Hydrogen, and the Clean Industrial Deal

Industrial sectors — particularly iron, steel, and cement — represent the hardest decarbonisation challenge. In the period 2000–2019, absolute greenhouse gas emissions from iron, steel, and cement grew more than in any other period in history. Emissions froze temporarily in 2014–2016 in the wake of the financial crisis, but returned to growth in 2017–2019 [RAG: CLIMATE_RAG — IPCC AR6 Chapter 11]. The structural challenge is that these sectors require either deep electrification, carbon capture and storage (CCS), or green hydrogen — none of which has yet achieved commercial scale in Europe.

Hydrogen offers a technically viable pathway for steelmaking: it can replace coal and coke as the reducing agent in iron production. The HYBRIT project in Sweden, piloted in 2020–2021, is among the most advanced real-world demonstrations of hydrogen-based steel decarbonisation [RAG: CLIMATE_RAG — IPCC AR6 Chapter 11]. However, electrolytic hydrogen produced in Europe currently represents only a small fraction of total hydrogen use for refining. The Netherlands "Holland Hydrogen" project represents one of the more developed downstream initiatives, yet supply and demand signals for

clean hydrogen in Europe remain insufficient to trigger large-scale infrastructure build-out [RAG: LNG_ENERGY_RAG — Oxford Institute for Energy Studies, Gas to Hydrogen 2024].

Material efficiency also offers significant mitigation potential. IPCC AR6 estimates that steel demand could be reduced by up to 40% through design optimisation, long-life construction, and high-quality recycling — reducing the volume of primary production that must be decarbonised in the first place [RAG: CLIMATE_RAG — IPCC AR6 Chapter 11]. Zero-carbon steel pathways are technically feasible but require integrated material efficiency, recycling, and production decarbonisation policies operating simultaneously.

The EU's policy response on the industrial side has evolved toward a "Clean Industrial Deal" framework. The OPEC World Oil Outlook 2025 notes that the EU's Clean Industrial Deal and the associated Affordable Energy Action Plan explicitly acknowledge the need to secure reliable energy supplies for European industries, including provisions for investments in overseas hydrocarbons infrastructure and long-term contracts — a pragmatic concession to energy security that sits in partial tension with Green Deal decarbonisation ambitions [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025].

4. Climate Adaptation: Flood, Heat, and Physical Risk Exposure

Mitigation policy dominates European climate discourse, but physical risk exposure is accelerating. IPCC AR6's climatic impact-driver mapping identifies Europe as facing rising confidence of increase across river flooding, heavy precipitation, pluvial flooding, extreme heat, hydrological drought, and agricultural drought [RAG: CLIMATE_RAG — IPCC AR6 Technical Summary]. Sea level rise and increased fire weather also feature in the European risk profile. Research on multi-model projections of river flood risk in Europe under global warming consistently finds rising losses: European flood losses have trended upward over the past 150 years, and early flood warning systems have been shown to yield significant monetary benefits in European contexts [RAG: CLIMATE_RAG — IPCC AR6 Chapter 13].

The IPCC AR6 chapter on European adaptation identifies key response clusters: efficiency improvements, water storage, and early warning systems for water scarcity; space reservation, ecosystem-based adaptation, and managed retreat for flooding; and nature-based solutions for heat alleviation — all of which are themselves under threat from continued warming [RAG: CLIMATE_RAG — IPCC AR6 Chapter 13]. Infrastructure is particularly exposed: high temperatures reduce electric transmission capacity while simultaneously increasing peak cooling demand, with "solar heat faults" in grid equipment documented under high-temperature, low-wind conditions.

Germany's national climate pathway — a 65% emissions reduction by 2030 relative to 1990 levels, followed by a 95% cut by 2050 through a complete switch to climate-neutral technologies, with residual emissions balanced through CCS — represents one of the EU's most quantified domestic trajectories [RAG: CLIMATE_RAG — IPCC AR6]. Achieving these targets will require not only physical infrastructure investment but institutional governance reform to coordinate adaptation at sub-national levels, particularly for regions where coal and carbon-intensive industries face managed transition [RAG: CLIMATE_RAG — IPCC AR6 Chapter 13].

5. Political Headwinds and the Frontrunner Paradox

Across the Atlantic, the US under the Trump administration withdrew from climate diplomacy forums and slashed multilateral climate funding, removing a counterpart whose engagement had underpinned the Paris Agreement architecture. This has amplified pressure on the EU to sustain the global momentum — but the EU itself faces mounting internal headwinds. Security challenges and a persistent rightward political shift within member states are testing the durability of climate ambition. The EU appears far more

stable than the US on climate policy, but even this historical frontrunner is under strain CNA[2025-08-29].

The UNDP's analysis of the trade-leakage dynamic underscores a structural problem for any unilateral climate actor: a country committed to reducing emissions may see higher sector carbon intensity imports from less regulated partners, offsetting domestic gains. Regional and global cooperation around nationally determined contributions is estimated to materially reduce the burden of the climate transition on lower-income populations — whereas unilateral action without coordination produces sub-optimal outcomes for both the environment and equity [RAG: UN_PRIMARY_RAG — UNDP Human Development Report 2023-24].

The UN governance framework — from UNEP's formation in 1972 through Rio 1992 and subsequent frameworks — has consistently struggled to translate multilateral ambition into binding national action. The EU's institutional architecture is more effective than international frameworks, but its external leverage depends on partners accepting the logic of carbon pricing. With the US retreating and key emerging markets resistant to CBAM framing as protectionism, the EU's frontrunner paradox deepens [RAG: GOVERNANCE_RAG — UN Environment and Climate Governance].

Outlook

The EU's 2030 climate architecture — centred on ETS expansion, CBAM phase-in, Fit for 55 legislative packages, and the Clean Industrial Deal — is the most comprehensive climate policy framework in existence. Execution, however, faces three compounding pressures. First, the ETS free allocation regime dilutes price signals precisely where they are most needed: in heavy industry. Second, CBAM's geopolitical friction with major trading partners risks regulatory escalation before it achieves its carbon-leakage correction function. Third, the rightward political shift across EU member states may constrain ambition in the 2024–2029 Commission cycle. On the physical risk side, Europe's adaptation gap is growing: flood, heat, and drought projections from AR6 point to rising damage costs even under moderate warming scenarios, and adaptation governance at sub-national level remains fragmented. The industrial hydrogen transition — critical for steel and chemicals — will not achieve commercial scale before 2030 without a step-change in green hydrogen supply infrastructure and demand-side policy certainty. The outlook is one of structural ambition under execution stress.

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